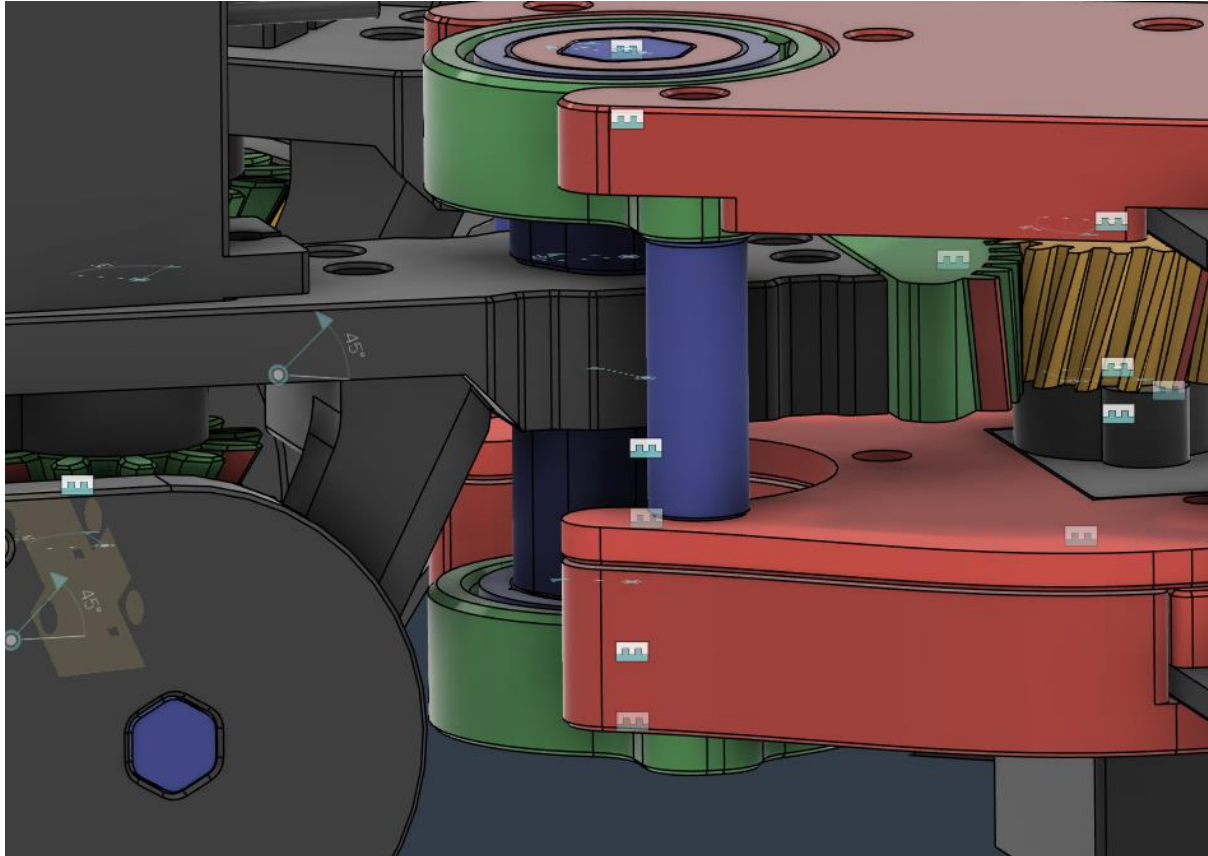


Model update 01.05.2025

In order to improve consistency, reduce reliance on precise tolerances, and reinforce the first joint, several changes have been made to the model.

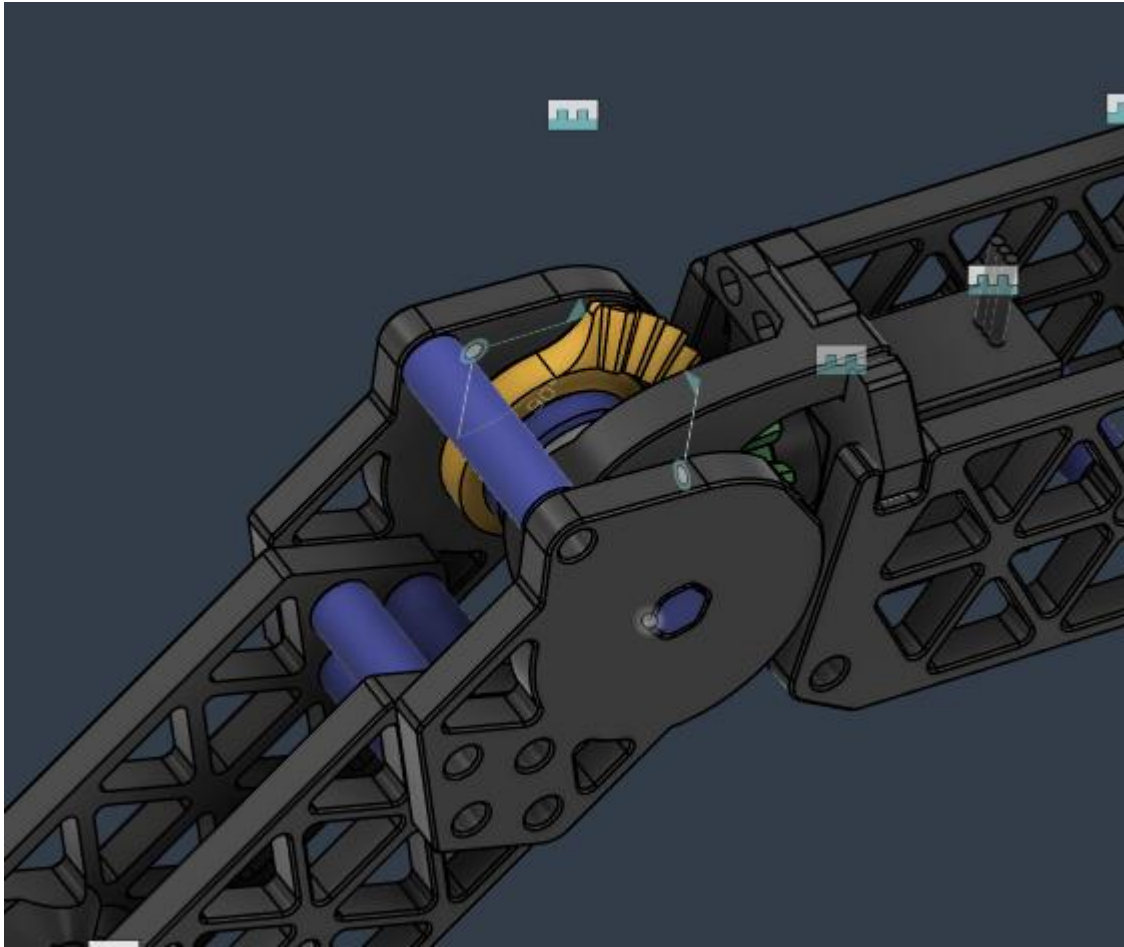
The impact that those changes have on the simulation is negligible.

Joint 0 changes:



The mounting bolt hole on the “up section/leg attachment/servo holder” has been relocated to ensure space for a new set of reinforcement rods. The rods prevent tension build-up from the torque (Y CCW for left legs of the hexapod) when the legs are lifted off the ground. Together with slots on the servo holder, the rods limit the motion preventing de-meshing of the gears.

Joint 3 changes:



New reinforcement rod has been added to ensure that the material won't bend outwards and skip a step on the gear assembly. This rod also prevents de-meshing of the gear assembly when a CCW rotation is applied past the range of motion.

Bug fixes:

Fixed missing holes on the platform mounting clips and bearing center hexagonal holes that were created due to an incorrect way that the Fusion handles downstream changes.

Misc:

Reduced the radius of the ball bearings mounting parts.

Changed how rods tolerance is calculated.

Numerous small changes with no downstream effect.